

Technical Skills Handbook

Python • Git • SQL • Linux • Data Structures & Algorithms

1. Python Programming

Python is widely used in automation, data science, AI, embedded applications and scripting. Learn variables, loops, functions, object-oriented programming, file handling, modules and virtual environments. Build small projects regularly to strengthen your fundamentals.

2. Git & GitHub

Git is a version control system used to track changes in code. Learn repositories, commits, branches, merging, pull requests and resolving merge conflicts. GitHub allows collaboration, portfolio building and deployment through GitHub Pages.

3. SQL & Databases

Understand relational databases, tables, keys, joins, filtering, aggregation and normalization. Practice writing SELECT, INSERT, UPDATE and DELETE queries. Learn basic database design and indexing concepts.

4. Linux & Command Line

Become comfortable using the terminal. Learn directory navigation, file permissions, package installation, shell scripting and process management. Linux skills are valuable for software engineering, cloud computing and embedded systems.

5. Data Structures & Algorithms

Study arrays, linked lists, stacks, queues, trees, graphs, hash tables and heaps. Learn algorithmic techniques including recursion, sorting, searching, dynamic programming and greedy algorithms. Practice coding problems consistently.

6. Learning Roadmap

Month 1: Python fundamentals and Git.

Month 2: SQL and Linux.

Month 3: Data Structures and Algorithms.

Month 4: Build portfolio projects and contribute to GitHub. Continue learning through real-world applications and open-source contributions.